

GENERAL AWARENESS [Field of Animal Science]

<u>Institute</u>	<u>Location</u>
1) IVRI - Indian Veterinary Research Institute	- Izatnagar, U.P
2) NDRI - National Dairy Research Institute	- Karnal, Haryana
3) CIRG - Central Institute for Research on Goat	- Mathdoom, U.P
4) CIRCA - Central Institute for Research on Cattle	- Meerut, U.P
[PDC] - Project Directorate on cattle	
5) CIRB - Central Institute for Research on Buffaloes	- Hissar, Haryana
6) CSWRI - Central Sheep & Wool Research Institute	- Ankanagar, Rajasthan
7) CARI - Central Avian Research Institute	- Izatnagar, U.P
8) NRCC - National Research Centre on Camel	- Bikaner, Rajasthan
*9) NRCE - National Research Centre on Equine	- Hissar, Haryana
10) NRCM - National Research Centre on Meat	- Hyderabad
*11) NRCMI - National Research Centre on Milkhun	- Jhapa, Nagaland
12) NRCP - National Research Centre on Pig	- Guwahati, Assam
13) NRCY - National Research Centre on Yak	- Dibrang, Arunachal Pradesh
14) NBAGR - National Bureau of Animal Genetic Resources	- Karnal, Haryana
15) DPR - Directorate of Poultry Research	- Hyderabad
[PDP] - Project Directorate on poultry	
*16) IRMA - Institute of Rural Management	- Anand, Gujarat
17) IGRI - Indian Grassland & Fodder Research Institute	- Jhansi, U.P
*18) NIANP - National Institute of Animal Nutrition & Physiology	- Bangalore
19) NIAH - National Institute of Animal Health	- U.P
20) NIAW - National Institute of Animal Welfare	- Faizabad, Haryana
21) Project Directorate on FMD	- Mukteshwar, Uttarakhand

Important Days :-

- * March 24 - World T.B Day
- * April 22 - World Earth Day
- * Last Saturday of April - World Veterinary Day
- * June 5 - World Environment Day
- * July 1st - Doctor's Day
- * July 6 - World Zoonosis Day
- * July 11 - World Population Day
- * July 16th - ICAR Foundation Day
- * September 28 - World Rabies Day
- * 2nd Friday of October - World Egg Day

Colour Revolutions in India

- | | | |
|---------------------|-----------------------------|-----------------------------|
| * Blue Revolution | - Fish production | - Prof. Hirshal Chaudhuri |
| * White Revolution | - Milk Production | - Dr. Verghese Kurien |
| * Green Revolution | - Food grains [Wheat, Rice] | - M.S. Swaminathan in India |
| * Silver Revolution | - Egg / poultry production | → Norman Borlaug in world |
| * Red Revolution | - Meat / Tomato production | |

First in India

- * Father of Veterinary science in India
(or)

SHALIHOTRA

First Veterinarian of India

- * First veterinary college in World - Lyon, France [1762]
- * First Veterinary college in India - Bombay Vet. College [1886]
- * First Veterinary University in India - TANUVAS, Tamil Nadu
- * First Embryo Transfer [ET] cattle calf in India - Annapurna Agr. University, Tirupattur 1986
- * World's First IVF Bull calf - PRATHAM at NDRI, Karnal, Haryana
- * World's First cloned Bull calf - SAMRUPA at NDRI, Karnal
- * World's First cloned Pashmina goat - NOORI by SKVAST - Kashmir

International Organizations :-

<u>Organisation</u>	<u>Year</u>	<u>Headquarter</u>	<u>Function</u>
1) World Trade Organisation [WTO]	1995	Geneva, Switzerland	To supervise & liberalize international Trade
2) International Organisation for Standardization [ISO]	1947	Geneva, Switzerland	International standards setting body
3) WHO	1948	Geneva, Switzerland	Concerned with international public health
4) FAO	1945	Rome, Italy	To defeat Hunger
5) Codex Alimentarius	1963	Rome, Italy	established by FAO & WHO
6) Office International des Epizooties [OIE]	1924	Paris, France	Supporting & promoting animal disease control.

New name :- WOAH

& OIE [World Organisation For Animal Health]

Rules & Regulation of Product Standards :-

- * Prevention of Food Adulteration Act, 1954 [PFA Act]
- * Meat Food Product Order [MFPO], 1973
- * Bureau of Indian Standard [BIS], 1986
- * Milk and Milk Product Order [MMPO], 1992
- * Food safety and Standard Act [FSSA], 2006

* PFA-1954 is replaced by FSSA-2006.

⇒ FSSAI :- Food Safety and Standards Authority of India

- * It is established under the FSSA act, 2006 to regulate food safety in India

* Headquarter - New Delhi

⇒ APEDA :- Agricultural and Processed Food Products Exports Dev. Auth

Headquarters - New Delhi

⇒ AWBI :- Animal Welfare Board of India

- * Established in 1962 Under PCA 1960

* Headquarter - Chennai

LIVESTOCK PRODUCTION AND MANAGEMENT

- * Livestock comprises of all farm animals having economic importance
[Milk / Meat / Wool / Draught] ex: Cattle, Buffalo, Savi, camel, pig, Horse, mule, Donkey, Pony etc.
- * Poultry, dog, cat and Rabbit are not fallen under Livestock category
- * 1 Livestock Unit = 1 adult Cattle / Buffalo
 = 2 Pigs
 = 3 calves
 = 5 sheep / Goat
- * Order - 1) Artiodactyla - Even-toed animals - eg: Ruminants, camel, Pig
 2) Perissodactyla - Odd-toed animals eg: Horse.

S.No.	Animal	Scientific Name	Chromosome No. [2N]
1)	Zebu cattle	Bos indicus	60 [Cattle, Yak, Goat]
2)	Exotic cattle	B. taurus	60
3)	River Buffalo / Water Buff.	Bubalus bubalis	50
4)	Swamp Buffalo	B. carabensis	48
5)	Goat	Capra hircus	60
6)	Sheep	Ovis aries	<u>54*</u>
7)	Yak	Bos grunniens	60
8)	Milkhun	Bos frontalis	<u>58*</u>
9)	Pig	Sus domesticus	38
10)	Cat	Felis domesticus	38
11)	Horse	Equus caballus	64
12)	Dog	Canis familiaris	78
13)	Chicken	Gallus domesticus	78
14)	Camel	Camellus dromedarius	<u>74*</u>
15)	Rabbit	Oryctolagus cuniculus	44

⇒ Common Laboratory Animals :

S.No.	Lab animal	Scientific Name	Gestation period
1)	Hamster	Mesocricetus auratus	16-18 d
2)	Mice	Mus musculus	20 days
3)	Rat	Rattus norvegicus	21 days
4)	Rabbit	Oryctolagus cuniculus	32-35 days
5)	Guinea pig	Cavia porcellus	59-72 days [65d]

⇒ Water Requirement of Domestic Animals

Animal	Water req. per day	Avg. daily amount of dung produced per animal
1) Buffalo	30-35 lit	25-40 kg
2) Dairy cattle	30-35 Lit	24 kg (18-34 kg)
3) Horse	36 Lit	13.50 kg
4) Pig	25 Lit	4 kg
5) Sheep & Goat	18 Lit	1.75 kg
6) Poultry	250ml	3 kg.

⇒ Water intake per kg Dry matter intake of various animals

Animal	Water per intake per kg DMI
1) Adult cattle	3-5 Lit
2) Milking cattle	3-5 Lit & Additional 4-5 lit / kg milk produced.
3) Suckling calf	6-7 Litres
4) Sheep & Goat	4 Litres
5) Swine	3 Litres
6) Poultry	2 Litres.

* Metabolic water comprises of 5-10% of total intake of the nutrient

⇒ Pseudo-ruminant :

* Stomach has 3 chambers [OMASUM absent]

* ex: Buffalo, cattle, sheep

Camel, llama, Alpaca, Vicuna

⇒ Bullock*: Castrated bull for draught purpose

⇒ Steer*: Castrated bull for Meat purpose

⇒ Heifer: A female cow above 1 year, which has not yet calved.

⇒ Cait / Caid / Runt* - Smallest piglet in a litter usually last to be borned.

⇒ Various Reproductive Stages of Cow:-

Name of period

No. of Days

Lactation period

305 days

Dry Period

60 days

Resting period

0 day

Calving interval

12 months [Buffalo - 13 months]

Gestation Period

283 days [Buffalo - 310 days]

Service time

85 days postparturient (2-3 months)

⇒ Buffalo Breeds

S.No.	Murrah group	Gujarat group	U.P group	South India group	Central India group
1)	Murrah	Jaffarabadi	Bhadawari	Toda	Nagpuri
2)	Nili-Ravi	Mehsana	Turai	South Kanara	Pandharpuri
3)	Kundi	Surti			Sambalpur
4)	Godavari				Kalahandi
					Jaxangi
					Mandla

* Nili-Ravi - Panchat Kalyani / Panchbhadra - Min. fat % in Buff. breeds (i.e. 4%)

* Heaviest Buff. breed - [Jaffarabadi]

* Wall eyes is peculiar character of Nili Ravi

* Bhadawari - Highest fat % (14%), Copper colour body coat, wedge shaped body

* Mehsana - Murrah x Surti, Longest lactation period, Black switch of the tail

* Most violent breed of Buffalo - [Toda]

* Sword shaped horns

* Dwarf breed of Buffalo - [Jaxangi] [Loissa]

- [Pandharpuri]

Animal	Terminologies related to various animals					House	Sound	Castrated Male
	Adult Male	Adult Female	Young Male	Young Female	New born	Group of Animals	Act of Mating	Act of Parturition
attle	Bull	Cow	Bull calf	Heifer calf	Calf	Head	Serving	Calving
sheep	Ram	Ewe	Ram Lamb	Gimmer lamb	Lamb	Flock	<u>Tupping</u>	Lambing
Goat	Buck	Doe/Nanny	<u>Buckling</u>	Goatling	Kid	<u>Trip/Band</u>	Serving	Kidding
Pig	Boar	Sow	<u>Boasting</u>	Gilt	Piglet	Drive*	<u>Coupling</u>	Farrowing
Horse	Stallion	Mare	<u>Colt</u>	<u>Filly</u>	Foal	Pack*	<u>Covering</u>	Foaling
Dog	Dog	Bitch			Pup	Kennel		Whelping
Cat	Tom	Queen			Kitten	<u>Clowder</u>		Queening
Camel	Maiya	Sand	Tosdia	Tosdi		Tola	Calving Lakhana	Calving
Buffalo	B. Bull	she Buffalo	B. bull calf	she buffalo	Calf	Head	Serving	Calving
Fowl (or) Chicken	Cock (or) Rooster	Hen (or) Pullet			chick	Flock		Hatching
Rabbit	Buck	Doe			kitten	colony		calving
Goose	Goander	Goose			Goosling			
Turkey	Tom	Hen			poult	Flour		

⇒ Cattle Breeds :-

Exotic :- Dairy breeds - Jersey, H.F, Brown swiss, Ayrshire, Guernsey
Beef breeds - Hereford, Angus, Shorthorn.

⇒ Jersey :- * Longest lactation period (365d) Among cattle
* Highest milk b.f among cattle [5.5%]
* Smallest exotic cattle breed.

⇒ H.F :- * Highest milk producer in the world
* Lowest milk b.f (3.5%)

⇒ Ayrshire :- * Most beautiful dairy breed

⇒ Brown swiss :- * Highest milk Lactose (5%)

* Jamaica Hope = 80% Jersey x 15% Sahiwal x 5% H.F

* Taylor = Shorthorn Bull x Native cow.

* First crossbred cattle developed in India - Taylor

Indian Cattle Breeds :-

1) Milch breeds :- Gir, Sahiwal, Red sindhi, Tharparkar

2) Draught :- Amritmahal, Khillari, Hallikar, Kangayam, Nagori, Malvi, Bachur, Bagur, Sirohi, Ponwar

3) Dual purpose :- Haryana, Kankrej, Rathi, Ongole, Deoni, Dangi, Mewali

⇒ Gir - * Longest lactation period among Desi breeds (325d)

⇒ Sahiwal - * Also known as Lola [Loose skin], Multani
* Best Indian milch breed.

⇒ Tharparkar * Highest Disease resistant
* Having colour change property

* Distinguishing feature b/w Red sindhi & Sahiwal is - Muzzle
Red sindhi - Dark coloured muzzle
Sahiwal - Light coloured.

- * Best Draught purpose breed of cattle in India - Amritmahal
- * Best Draught purpose breed of Rajasthan - Nagori
- * Best Draught Purpose breed of Maharashtra - Khillari
- * Smallest cattle breed in the world - Vechur [Kottayam, Kerala]
- * Pride of Tamilnadu - Kangayam
- * Hilly breeds of cattle - Siri, Ponwar & Badri
- * Best Dual purpose cattle Breed - Hariana
- * Rathi - Red sindhi x Tharparkar x Sahiwal
- * Heaviest Indian cattle breed & Famous Sawai-Chal gait - Kankrej
- * Cattle with pure white colour - Ongole
- * Mewati - Gir x Rathi x Nagori
- * Nimari - Gir x Khillari
- * Deoni - Gir x Dangi x local animal
- * Badri - Newly registered breed of Uttarakhand also known as Pahadi cow / Hilly cow

⇒ SHEEP BREEDS :

Exotic sheep :-

- 1) Fine wool breeds - Merino, Rambouillet
 - 2) Long coarse wool breeds - Lincoln, Leicester
 - 3) Fur breeds - Karakul
 - 4) Mutton breeds - Southdown, Down, Southdown
- * Best fine wool breed in the world - Merino
 - * Largest & Heaviest sheep breed in world - Lincoln
 - * Best pelt (good quality fur) producing sheep breed in world - Karakul
 - * Best Mutton breed in world - Southdown

⇒ Indian Sheep Breeds :

- * Indian Marino - Chokla*
- * Canary / yellow coloration of wool is seen in - Nali
- * Highest disease resistant & Triple purpose Sheep Breed - Maxwasi*
- * Yellow streak above the eyebrow - Pugal
- * Nellore - * Tallest breed of sheep in India
 - * Goat like sheep breed [No wool]
 - * Best Mutton breed of India
- * Smallest sheep breed - Mandya
- * Most prolific sheep breed of India - Garole
- * Biggest sheep among Kashmir breeds - Gurez
- * Gaddi breed is used for making Kullu shawls & Blankets

⇒ GOAT BREEDS :

Exotic breeds : Sannen, Alpine, Anglo-Nubian, Angora, Toggenberg, osenberg, Boer.

- * Milk queen of Goat world - Sannen
- * Angora - * Mohair producing breed of goat
 - * Sheep like goat
- * Anglo-Nubian - * Jersey of Goat
 - * Most outstanding Dual purpose breed
 - * Widely used for upgrading of desi stock

Indian breeds :

- * Jamunapari - * Biggest & most majestic & most beautiful breed of India
 - * Roman nose, Creamy white body colour with shining hair
- * Origin of Barbari Goat - East Africa*
- * Dwarf breed suitable for stall feeding
- * City breed.
- * Indian goat with maximum milk hf [5%]

* Black Bengal - * Highly prolific goat breed of India

* Best chevon breed of India

* Pashmina type - Chegu & Changthangi

* Changthangi also known as Pashmina

⇒ Pig BREEDS :-

Indian Pig breeds :-

- 1) Ghoongroo - W.B
- 2) Niang Megha - Meghalaya
- 3) Agonda goan - Goa
- 4) Tenyi Vo - Nagaland
- 5) Nicobari - Andaman & Nicobar
- 6) Doom - Assam

Exotic breeds :-

- 1) England breeds - LWH, MWH, Berkshire, Tamworth
- 2) Denmark breeds - Landrace
- 3) U.S.A breeds - Duroc, Chester-White, Poland-china

* Heaviest breed of pig - LWH

* Excellent Pork breed - MWH

* Tamworth - Golden red colour body

Tall breed known for lean meat

* Highest quality bacon in the world - Landrace

* Landrace is white colour breed with some black skin spots - Freckles

⇒ HORSE BREEDS :-

* Kalthiawari, Manipuri, Marwari, Spiti, Zanskari, Bhutia

* Best breed of Horse - Kalthiawari

* Horse breed used for polo, military & racing - Manipuri

Best mutton breed - Nellore

Best chevon breed - Black Bengal

Prolific goat - Black Bengal

Prolific sheep - Gasole

Tallest sheep - Nellore

Smallest sheep - Mandya

⇒ Camel Breeds :

- 1) Breeds of Rajasthan — Bikaneri, Jaisalmeri, Marwari, Mewati, Mewari, Talori
- 2) Gujarat — Kutchi, Khari
- 3) M.P — Malvi

* Best camel breed — Bikaneri

* Khari — also called as Swimming camel

Tolerant to high saline water

* Indian camels are 2 types
 Plain camels
 Hill camels
 Reversine camels — Baggage type
 Desert camels — Riding type
 Baggage type

* Desert camel breeds — Bikaneri & Jaisalmeri

Breed statistics :

* Total no. of indigenous breeds in country — 184*

* No. of Cattle Breeds — 43
Bulalo — 16 * Donkey — 2
Sheep — 43 * Yalk — 1
Goat — 34
Horses & Ponies — 7
Camel — 9
Pig — 8

Chicken — 19, Duck — 1, Geese — 1

⇒ Newly Registered breeds — 5th Sep 2018 15 New breeds

	<u>New breed</u>	<u>Home tract</u>	<u>Home tract</u>
1) <u>Cattle</u> :	1) <u>Ladakhi</u> -	J&K	6) <u>Goat</u> - 1) <u>Kahmi</u> -
	2) <u>Konkan Kapila</u> -	Maharashtra & Goa	2) <u>Rohilkhandi</u> -
2) <u>Bulalo</u> :	1) <u>Luit (swamp)</u> -	Assam & Manipur	3) <u>Assam Hill</u> -
	2) <u>Bugut</u> -	Tamil Nadu	4) <u>Bidri</u> -
	3) <u>Chattisgarhi</u> -	Chattisgarh	5) <u>Nandidurg</u> -
3) <u>Sheep</u> :	1) <u>Panchali</u> -	Gujarat	6) <u>Bhakawali</u> -
4) <u>Pig</u> :	1) <u>Ghuxrah</u> -	U.P	7) <u>Donkey</u> - 1) <u>Halari</u> -
5) <u>Chicken</u> :	1) <u>Uttara</u> -	Uttarakhand	

Milking Methods :-

- A) Hand milking - Common practice among individual farmers
- B) Machine milking - Common Practice in an organised farm

⇒ Hand milking - 3 types

- 1) Full hand Milking / Fisting - Best method of milking dairy animals
- 2) Knuckling - Wrong method of milking
- 3) Stripping - Mainly used in sheep & goat
To remove the residual milk

* Hindquarters of the udder are slightly larger than the front ones and contain more milk

* Ratio of milk obtained from ⁶⁰Hind & ⁴⁰forequarters in cattle - 60:40

Methods of Drying a Cow :-

- 1) Incomplete milking - Best method of drying off a cow
- 2) Intermittent milking -
- 3) Complete cessation - Wrong method

Methods of Age Determination :-

- 1) By record keeping
- 2) By physical appearance
- 3) By counting Hoan rings Age = N + 2 * N = no. of hoan rings

* In cattle first hoan ring appears at the age of 3 years

- 4) By dentition - most commonly used method for age determination

* Age of the animal determined by incisors of lower jaw

* Gummer - an animal that has lost its all teeth

* Gimmer - sheep b/w 1st & 2nd shearing

* Canine teeth are absent in Ruminants & Female Horse (mare)

* Incisors of Horse - Dental star - mark on central pulp cavity of incisors
Galvayness groove - Groove in upper corner incisor
10 years

Dental Formulae of Various Animals

<u>Animal</u>	<u>Dental formula</u>	
1) Ruminants	$\frac{0}{4}, \frac{0}{0}, \frac{3}{3}, \frac{3}{3} = 32$	* Upper jaw - Dental pad * Canines are absent in all ruminants
2) Pig	$\frac{3}{3}, \frac{1}{1}, \frac{4}{4}, \frac{3}{3} = 44$	
3) Horse - stallion	$\frac{3}{3}, \frac{1}{1}, \frac{3}{3}, \frac{3}{3} = 40$	
Mare	$\frac{3}{3}, \frac{0}{0}, \frac{3}{3}, \frac{3}{3} = 36$	* <u>Canines are absent</u> *
4) Dog	$\frac{3}{3}, \frac{1}{1}, \frac{4}{4}, \frac{2}{3} = 42$	
5) Cat	$\frac{3}{3}, \frac{1}{1}, \frac{3}{2}, \frac{1}{1} = 30$	
6) Camel	$\frac{1}{3}, \frac{1}{1}, \frac{3}{2}, \frac{3}{3} = 34$	

* Full mouth condition :- Age at which all permanent teeth appear

Cattle, Buffalo & Goat	- 5 years
Horse	- 4.5 years
Sheep	- 4 years
Camel	- 7 years

⇒ Methods of identification of animals & poultry :-

Permanent methods

Branding
Tattooing
Ear notching [Pigs]

Temporary methods

Ear tagging
Hoof marking [Horse]
Horn marking [Cattle]
Wool marking [Sheep]
Wing tagging [Poultry]
Neck chain [Dog]

* Branding - most suitable for cattle, buffaloes, camels & horses

* Lower part of the thigh - best site for branding

* Tattooing - Best method for sheep, goat & newborn calves

* site - inside the ear b/w large veins

* Ear notching commonly used in pigs

⇒ Avg. Life span & Age at puberty :

<u>Species</u>	<u>Avg. Life span</u>	<u>Puberty age in months</u>
1) Camel	40 yrs	3-4 years
2) Horse	20 years	18 months (1.5 yrs)
3) Cattle	20 years	18-24 M
4) Buffalo	20-25 yrs	18-24 M
5) Goat	15 yrs	7-10 M
6) Sheep	12 yrs	7-10 M
7) Swine	8 yrs	6-7 M
8) Canine	10 yrs	7-10 M.

⇒ Age of castration & Weaning :

<u>Animal</u>	<u>Age of castration</u>	<u>Age of Weaning</u>
1) Calf	8-10 Wks	90 days (3 Months)
2) Lamb	2 Wks	14 Wks
3) Kid	2-4 Wks	16 Wks
4) Piglet	1 Wk	8 Wks

* Castration in female - spaying / ovariectomy

* Castrated bowl is known as Capon & process is called caponisation

* Castrated Tom / male cat - Neuter & process is Neutering.

⇒ Dehorning / Disbudding :-

* Best Age of dehorning in calf - Before 10 days old.

Methods :-

- 1) Chemical method - KOH (caustic Potash) is used
- 2) Electrical method - Electric dehorner @ 1000°F / 10 sec
- 3) Mechanical -

⇒ Differences b/w Riverine & Swamp Buffalo :-

Water buffalo - *Bubalus bubalis*
2N = 50

Swamp buffalo - *Bubalus carabensis*
2N = 48

Type → Dairy breeds

→ Draught & Meat breeds

Native → India, Pak, Arab

→ Philippines, China, Burma

* 2 swamp buff. breeds in India - Manipuri & Assamese

Manipuri type - True swamp buffalo

Animal	Generation interval - Avg. age of parents when their offspring are born
1) Buffalo	4.5 years
2) Cattle	4 years
3) S&G	3 years
4) Swine	2.5 years
5) Poultry	0.65 years

- * Thermo-neutral zone for pig & cattle - $20-26^{\circ}\text{C}$
- * Thermo-neutral zone for poultry - $13-24^{\circ}\text{C}$
- * Physiological zero temp. for poultry - 21.1°C
- * Comfort zone for pig & cattle - $21-24^{\circ}\text{C}$
- * Comfort zone for poultry - $18-21^{\circ}\text{C}$
- * Flushing - Feeding of extra concentrate 2-3 wks prior to breeding
- * Steaming up - Feeding of extra concentrate in last 6-8 wks of pregnancy
- * Eyeing - Removal of wool around eyes to prevent wool blindness
- * Ringing - Clipping of wool at the region of penis
- * Calving interval = $\frac{\text{Lactation period} + \text{Dry period}}{\text{Service period} + \text{Gestation period}}$
- * Calving interval - interval b/w 2 successive calving of an animal
- * Calving index - Avg. interval b/w 2 successive calving of a herd.
- * Cattle has 10 times more no. of sweat glands than Buffalo.
- * Prolificacy - Ability to give birth to large no. of young ones
- Prolific sheep breed - Garole
- Goat - Black bengal
- * First faeces voided by a newborn calf - Meconium
- * Best method of carcass disposal - Burial method
- * Indubrasil - Gir x Kankrej
- * A cattle breed from Tamilnadu, used for Jellicyt during festival - Pullikulan
- * Annual culling rate in Dairy unit is - 25%
- * Pelt production from Karakul breed - very young lamb & betal lamb.

- * Ship of snow - Yak
- * Ceremonial ox - Milthun
- * Ship of Desert - Camel
- * Poor man's cow - Goat
- * In cattle rearing (dairy) - Feed cost alone account for 60-65% of total breu expenses
- * Pig breeding - Feed cost - 80% of total expenses
- * Initiation of Lactation is called - Lactogenesis
- * Maintenance of Lactation - Galactopoiesis
- * Skin of unborn calf is called - Slunk
- * Browsers - Goat & Camel

Cattle Housing System :-

2 types

- 1) Loose Housing System
- 2) Conventional Dairy Barn / Stanchion dairy Barn

Requiriments of Loose Housing System :-

Type of animal	Floor space requirement (m ²)		Max. no. of anls per pen	Height of shed in cm at Eaves
	Covered Area	Open area		
1) Bulls	12 m ²	120 m ²	1	175 cm in medium & heavy rainfall areas & 220 in semi arid & arid areas.
2) Cows	3.5	7	50	
3) Buffaloes	4	8	50	
4) Young calves	1	2	30	

⇒ Ratio of open area : covered area in case of loose housing system 2:1

⇒ Cow shed arrangement :-

- 1) Single row - If herd is small (Less than 10 animals)
- 2) Double row - If herd is large (More than 10 animals)

Double row housing :-

1) Head to Head system

* Feed distribution is easy

2) Tail to Tail system

* Cleaning & Milking is easy

* Lesser danger of disease spread

* Desirable slope of floor in animal house should have 1:40 gradient

* Age of castration in bull — 1 year

Age of maturity of bull — 2.5-3 years

* Skin of unborn calf — Skunk

* Pig skin is called — Rind

* Large animal skin — Hide

* Only sheep breed of south India producing Fine Wool — Nilgiri*

* The internal width of the shed [from wall to wall] is called — Span*

* Calving interval* = service period + gestation period.

* Better use of productive resources seen in — Diversified breeding

* Wool quality can genetically improved by selecting animals for Reduced medullation % [Good quality Wool — Less (or) No medulla]

* Wet : Dry ratio in organized dairy herd — 2:1

* Slope of the roof in animal shed should not exceed — 45°*

⇒ Milk contribution :-

Buffalo — 54%

Cow — 42%

Goat — 29%

* Oldest domesticated animal — Dog

Last domesticated animal — Horse

* Indian cattle maintain the temp. through — Dewlap

SWINE MANAGEMENT :-

- * FCR of pig — 1:3
- * Ca is deficient in meat of pig.
- * Pork is rich in Phosphorus, Iron & have high energy value due to more fat & lower water content.
- * Puberty age of pigs — 6-8 months [^{b.wt} 68 kg]
- * Breeding age of Gilts — 10-12 months
- * Maturity age of Boar — 18-24 months
- * Age of castration in pig — 4-6 wks
- * Age of Weaning piglets — 8 weeks
- * Test for detecting sow in heat — Riding test [Immobilisation of sow in heat along with erection ears when manual pressure applied on back]
- * Energy required is expressed as amount of DE/ME per kg of daily ration
- * Pig ration should must have — More concentrates
Less roughages
- * Bacon meat — From back & sides of a pig which is preserved by curing with brine.
- * Castrated male pig — Barrow/stag Barrow - Before sexual maturity
stag - After puberty
- * Group of pigs called — Drove
- * Hog — uncastrated male pig above 55kg b.wt
- * Lard — Fat from pig carcass after it has been tendered.
- * Sty — House of pig
- * Piggery — Place where pigs are kept
- * 4 pairs of needle like teeth are present in newly born piglets
- * Mortality of piglets — From birth to weaning — 25-30%
Post weaning — 3%
- * No. of teats in sow — 6-12

Swine Dentition :-

- * Temporary dentition - $\frac{3}{3}, \frac{1}{1}, \frac{3}{3}, \frac{0}{0} = 28$
- * Permanent - $\frac{3}{3}, \frac{1}{1}, \frac{4}{4}, \frac{3}{3} = 44$
- * Temporary teeth are completed by 5th month
- * After 5th month, these are start to replaced by permanent teeth
- * 6 month - corner permanent incisor
- 10 month - central permanent incisor
- 12 month - Intermediate permanent incisor
- 1.5 years - All permanent teeth
- * Avg. Life span of a Pig - 6-8 years

SHEEP MANAGEMENT :-

- Cross breeds of sheep :-
- 1) Hissardale - Australian Merino x Bikaner/Magra (ewe)
(Ram)
 - 2) Arivastara - Rambouillet x Chokla
 - 3) Avikalin - Rambouillet x Malpura
 - 4) Kashmir Merino - Merino x Gaddi/Bhakrawal/Poonchi
 - 5) Nilgiri - Coimbatore x Tosmanian Merino
 - 6) Coarsedale - Merino ewes x Lincoln ram

- * Flushing :- Feeding extra nutrients 2-3 weeks prior to breeding season
- * Crutching - shearing of wool around tail
- * Tagging - shearing of wool & dirt from dock region
- * Eyeing - clipping of wool around eyes
- * Ringling - Removal of wool region around penis
- * Limiting amino acid for sheep - Methionine
- * DM req. for sheep - 2.5-3 kg/day

Management of sheep

Season

Winter

Spring

Summer

Rainy

Purpose

Supplemented breeding

Mating

Gestation season

Lambing season

- * only milk producing sheep - Sonadi
- * Canary yellow colour wool - Nali
- * Kemp - chalky, white, lustreless, coarse hair like fibre
- * Lanolin - Waxy & greasy substance secreted from sheep skin
- * Indian Merino - Chokla
- * True Wool fibre / Fine wool does not contain the medulla

Fine wool — No medullation* eg:- Merino, chokla*

Carpet wool — 10-20% Medullation eg:- Magra / Bikaneri